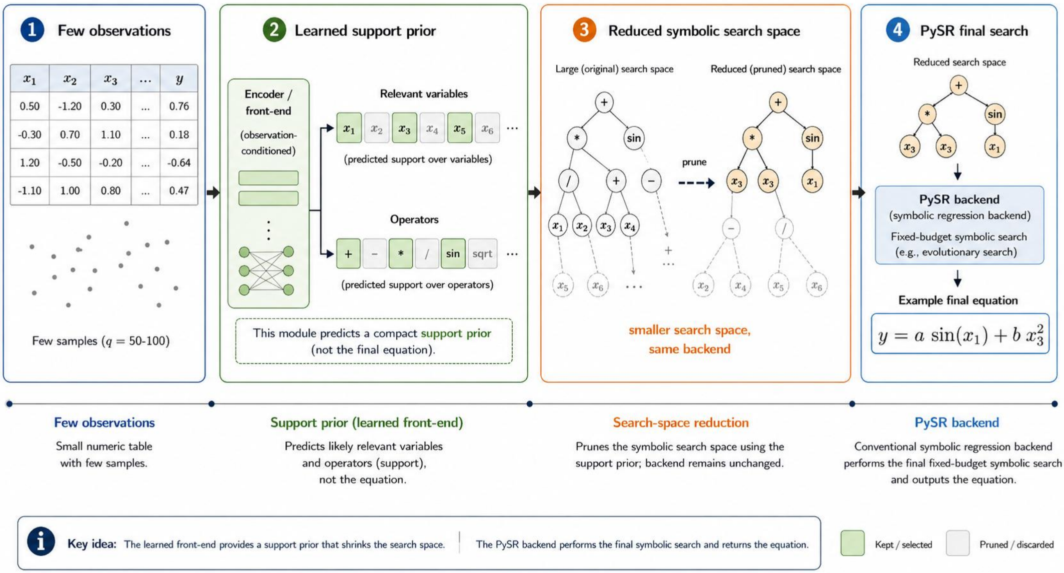

1 Few observations

x_1	x_2	x_3	...	y
0.50	-1.20	0.30	...	0.76
-0.30	0.70	1.10	...	0.18
1.20	-0.50	-0.20	...	-0.64
-1.10	1.00	0.80	...	0.47

Few samples (q = 50-100)
2 Learned support prior
Encoder / front-end
(observation-conditioned)
Relevant variables
x_1 x_2 x_3 x_4 x_5 x_6 ...
(predicted support over variables)
Operators
+ − * / sin sqrt ...
(predicted support over operators)
This module predicts a compact support prior (not the final equation).
3 Reduced symbolic search space
Large (original) search space
Reduced (pruned) search space
prune
smaller search space, same backend
4 PySR final search
Reduced search space
PySR backend
(symbolic regression backend)
Fixed-budget symbolic search (e.g., evolutionary search)
Example final equation
$y = a\,\sin(x_1) + b\,x_3^2$
Few observations
Small numeric table with few samples.
Support prior (learned front-end)
Predicts likely relevant variables and operators (support), not the equation.
Search-space reduction
Prunes the symbolic search space using the support prior; backend remains unchanged.
PySR backend
Conventional symbolic regression backend performs the final fixed-budget symbolic search and outputs the equation.
Key idea: The learned front-end provides a support prior that shrinks the search space.
The PySR backend performs the final symbolic search and returns the equation.
Kept / selected
Pruned / discarded